THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(c)** u / sqrt(2)	v squared = u squared - 2g(h/2) with h = u squared/2g.
6	**(c)** 1 : 4	a is inversely proportional to m.
7	**(d)** 100 J	Half converted to PE.
8	**(b)** 0.68 m and 3 m	lambda = v/f.
9	**(a)** 8 km/h	24 km/ 3 h.
10	**(a)** the pull of the book on the Earth	Third-law pair.
11	**(b)** 250 J	mgh.
12	**(b)** 24 m/s	0.6 x 40.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(a)** 4 m/s squared	F/m.
15	**(b)** 480 W	2400/5.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Harmonic mean 2 x 30 x 60/90 (1); = 40 km/h (1).	1+1
17	a = -2 m/s squared (1); t = 5 s, s = 25 m (1).	1+1
18	KE change = (1/2)(2)(25 - 9) (1); = 16 J (1).	1+1
19	Distance = 170 m (1); v = 170/0.5 = 340 m/s (1).	1+1
20	Dropped: 2 s (1); Thrown down: 20 = 10t + 5t squared gives t = 1.24 s (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	v = 20 m/s; s1 = 50 m (1); s2 = 200 m (1); t3 = 10 s; s3 = 100 m; total = 350 m (1).	1+1+1
22	a = 4 m/s squared (1); F = 6000 N, s = 50 m (1); Weight = 15000 N; force is 0.4 of weight (1).	1+1+1
23	PE lost = 200 J (1); KE gained = 90 J (1); Work against friction = 110 J (1).	1+1+1
24	Wavelengths 4 : 1 (1); Speeds equal (1); Loudness equal; pitch higher for 400 Hz (1).	1+1+1
	OR: s1 = a/2 = 2, so a = 4 m/s squared (1); s4 = 0.5 x 4 x 16 (1); = 32 m (1).	1+1+1
25	a = v squared/2s = 2 x 10^6 m/s squared (1); F = 0.01 x 2 x 10^6 = 20000 N (1); t = v/a = 0.0002 s (1).	1+1+1
	OR: Energy proportional to x squared (1); Doubling the extension gives 4 times the energy (1); 16 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Distances inversely proportional to a: 2 km and 1 km (1); v squared = 2 x 0.5 x 2000, v = 44.7 m/s (1); t1 = 89.4 s (1); t2 = 44.7 s; total 134 s (1); Check s2 = 44.7 squared/2 = 1000 m (1).	1+1+1+1+1

	OR: a = 3 m/s squared (1); v = 15 m/s (1); s = 37.5 m (1); p = 120 kg m/s (1); F = change in p/t = 24 N (1).	1+1+1+1+1
27	v = sqrt(2 x 10 x 3.2) = 8 m/s (1); KE at bottom = 64 J (1); Work by friction = -64 J (1); f x 4 = 64, f = 16 N (1); Energy conserved: PE to KE to heat (1).	1+1+1+1+1
	OR: lambda = 0.782 m (1); t = 344/344 = 1 s (1); Waves between source and wall = 172/0.782 = 220 (1); Echo path 344 m: 440 waves (1); Frequency equals the waves per second (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 150 m	1500 x 0.2/2.
2	**(c)** 1.5 t	v = at; retard time = at/2a = t/2.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 4.8 km/h	Harmonic mean.
7	**(c)** 20 m/s	a = 4 m/s squared; v = 20 m/s.
8	**(b)** 3h/4	KE = 1/4 of mgh.
9	**(b)** 2 s	680/340.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** reducing the friction by rollers	Reduce friction.
12	**(b)** 4 m/s	sqrt(2 x 10 x 0.8).
13	**(b)** 86 m	vt.
14	**(b)** 2t	t proportional to sqrt(h).
15	**(c)** 200 N	0.1 x 20 / 0.01.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Normal force from the ground balances the weight (1); Net force zero (1).	1+1
17	P = F v (1); F = 10000 N (1).	1+1
18	Different harmonic content (1); Waveform shapes differ (1).	1+1
19	a = v squared / 2s (1); = 225/120 = 1.875 m/s squared (1).	1+1
20	Net force zero at constant velocity (1); Hence pull equals friction (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	T = 0.00143 s (1); lambda = 0.486 m (1); Waves in 5 s = 3500 (1).	1+1+1
22	v = 8 m/s; s1 = 16 m (1); s2 = 48 m (1); t3 = 2 s; s3 = 8 m; total 72 m (1).	1+1+1
23	F = 10/0.01 = 1000 N (1); Third law: nail exerts 1000 N on the hammer (1); Force on the nail 1000 N (1).	1+1+1
24	By force = 90 J (1); By gravity = -60 J (1); KE = 30 J, v = 5.5 m/s (1).	1+1+1
	OR: Reflections from the walls prolong it (1); Reverberation (1); No reflections in the open (1).	1+1+1
25	Relative speed = 18 km/h = 5 m/s (1); Relative distance = 100 m (1); Time = 20 s (1).	1+1+1
	OR: a = 5 m/s squared (1); v = 20 m/s, change in momentum 120 kg m/s (1); s = 40 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	T = 2 x 3.14/4 = 1.57 s (1); Speed constant 4 m/s (1); Velocity changes direction (1); Displacement in half period 2 m (1); Average velocity = 2/0.785 = 2.55 m/s (1).	1+1+1+1+1

	OR: Change = 0.16 x 70 = 11.2 kg m/s (1); Impulse = 11.2 N s (1); F = 560 N (1); Doubling time halves force (1); Same impulse (1).	1+1+1+1+1
27	3000 kW x 24 = 72000 kWh (1); Cost = 3.6 lakh rupees (1); Number of bulbs = 3 x 10^6/100 (1); = 30000 bulbs (1); Energy = P x t (1).	1+1+1+1+1
	OR: Time gap less than 0.1 s in a small room (1); A big hall delays reflections (1); Use of absorbers (1); Curved ceilings and sound boards (1); Ceiling height above 17 m avoided or treated (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 3 : 20	0.3 : 2.
2	**(b)** 340 m/s	34/0.1 = 340 - note: for illustration.
3	**(a)** 24 m/s	2 x 20 x 30/50.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(c)** A is true but R is false.	A is true; R is false: KE is zero there.
6	**(d)** A is false but R is true.	A is false (loudness does not depend on frequency); R is true.
7	**(d)** A is false but R is true.	A is false: acceleration is g throughout; R is true.
8	**(a)** 10 m/s squared downward	Only gravity acts.
9	**(b)** 4 : 1	KE = p squared/2m.
10	**(c)** 1 : 3	Distance proportional to time.
11	**(a)** 1 : 3 : 5	Odd numbers.
12	**(a)** the same for both	F = change in p/t.
13	**(a)** 72 J	F = m(g + a) = 24 N.
14	**(d)** 20 m	t1 = sqrt(2h/g), t2 = h/v: h = 20 gives 2 + 0.06.
15	**(a)** 31.25 m	t up = 2.5; h = 0.5 x 10 x 6.25.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Work against gravity depends only on the vertical rise (1); PE = mgh (1).	1+1
17	Sound travels faster in steel (1); It reaches the ear earlier (1).	1+1
18	u = g x 2 = 20 m/s (1); H = 20 m (1).	1+1
19	Same direction: 14/2 = 7 m/s squared (1); Opposite: 4/2 = 2 m/s squared (1).	1+1
20	PE at 8 m = 80 J (1); Energy lost = 100 - 80 = 20 J (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Total mass 80 kg (1); W = 4000 J (1); P = 400 W (1).	1+1+1
22	d1 = 340 m (1); d2 = 510 m (1); Distance between cliffs = 850 m (1).	1+1+1
23	t = 1 s, v = 10 m/s (1); a = 10/0.1 = 100 m/s squared (1); Bending knees increases the stopping time (1).	1+1+1
24	At 40 N: net force zero, constant velocity or rest (1); At 60 N: net 20 N (1); Box accelerates in the direction of the push (1).	1+1+1
	OR: Useful P = 3000 W (1); Input = 3000/0.6 (1); = 5000 W (1).	1+1+1
25	Initial distance 510 m (1); New distance 425 m (1); New echo time 2.5 s (1).	1+1+1
	OR: Closing speed with truck 45 m/s (1); In 6 s: 270 m covered (1); 270 < 400, so safe (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Initial velocity 10 m/s upward (1); -75 = 10t - 5t squared (1); t squared - 2t - 15 = 0, t = 5 s (1); v = 10 - 50 = -40 m/s (1); Speed 40 m/s downward (1).	1+1+1+1+1
	OR: Momentum before firing = 0 (1); Momentum of bullet = 0.02 x 200 = 4 kg m/s (1); Gun has equal momentum in the opposite direction (1); Recoil velocity = 4/4 = 1 m/s opposite to the bullet (1); Law of conservation of momentum (1).	1+1+1+1+1

27	(a) 15 m/s (1); (b) 135000 J (1); Distance 75 m; resistive work 45000 J (1); (c) Engine work 180000 J (d) 18000 W (1); (e) Force = 1200 x 1.5 + 600 = 2400 N; P = 36000 W (1).	1+1+1+1+1
	OR: Observer sees flash/clap and starts stopwatch (1); Stops when the sound is heard (1); v = 1020/3 = 340 m/s (1); Repeat and take the mean (1); Precautions: large distance, wind, reaction time (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND